

Actors / External Entity:

Customer

Objects / Lifelines:

Shopping App

Product Catalog

Purchase Service

Checkout System

Payment Gateway

Messages / Interactions:

1. Browse Items
2. Query Items / Item
3. Item Details
4. Display Items
5. Select Item
6. Display Selected Item & Price Info
7. Purchase Item
8. Create Purchase / Order
9. Checkout Request
10. Authorize Payment
11. Payment Approved
12. Order Confirmed
13. Purchase Success
14. Display Confirmation (Order Details)

Relationships / Message Flow:

Customer → Shopping App

Shopping App → Product Catalog

Product Catalog → Shopping App

Shopping App → Customer

Customer → Shopping App

Shopping App → Customer

Customer → Shopping App

Shopping App → Purchase Service

Purchase Service → Checkout System

Checkout System → Payment Gateway

Payment Gateway → Checkout System

Checkout System → Purchase Service

Purchase Service → Shopping App

Shopping App → Customer

UML Components Present:

- Actor
- Objects/Lifelines
- Activation Bars
- Message Arrows
- Return Messages (dashed arrows)
- Sequence Flow (top to bottom)
- Interaction Modeling

Online Shopping System – Sequence Diagram

